

Problem-Solution Fit

Date	01 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] • Farmers • Agricultural Researchers • Policymakers 	6. CUSTOMER LIMITATIONS [CL] • Limited budget • Limited internet access • Limited technical knowledge 	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Existing Solutions: Traditional farming advice, manual soil testing, basic apps. Pros: • Easy to access • Low initial cost Cons: • Not data-driven • Less accurate • Time consuming 
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Wrong crop selection • Low yield • Inefficient resource usage • High risk due to unpredictable weather & soil conditions 	9. PROBLEM ROOT / CAUSE [RC] Lack of data-driven crop recommendations and environmental analysis. 	7. BEHAVIOR + ITS INTENSITY [BE] • Frequently checks weather and soil conditions before sowing. • Seeks best practices to improve productivity. • Open to adopting technology for better outcomes. 
3. TRIGGERS TO ACT [TR] • Need to improve productivity and increase yield • Need to reduce farming risks • Changing weather patterns • Increasing input costs 	10. YOUR SOLUTION [SL] AI/ML-based OptiCrop system recommending suitable crops using NPK, pH, rainfall, humidity and temperature. 	8. CHANNELS OF BEHAVIOR [CH] Online: • Mobile app • Website Offline: • Agriculture offices • Field visits 
4. EMOTIONS BEFORE / AFTER [EM] Before: Confused, uncertain After: Confident, satisfied 		